STATISTICS · SAMPLING AND EXPERIMENTATION

Sampling Methods Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Sampling Methods Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A reporter asks the first 16 students who walk in the front door how much they slept. Will the average be accurate? Explain.
   Probably not: they are not a random sample, so the method can be biased. (In the lab, front seats are 12th graders, who sleep less, so it is too low.)
2. Does a larger sample fix a biased sampling method?
   No. It reduces variability, not bias.

2 Explore use the tool

1. Take three simple random samples (n = 16) one at a time. Record each $\bar{x}$. Did any equal $\mu = 7.32$ exactly?
   Answers vary; usually none exactly. Random samples vary around μ.
2. Convenience sample, n = 16: take 200+ samples. Record the center and the SD of $\bar{x}$. Explain both.
   Center ≈ 6.66, SD = 0: every convenience sample is the same 16 students, so it is biased low and never varies.
3. Convenience sample, n = 64: record the center. Did the bias go away?
   ≈ 6.80, still well below 7.32. The bias remains.
4. With n = 16, compare 200+ SRS samples with 200+ stratified samples. Which is less variable? Why?
   Stratified (SD ≈ 0.145 vs ≈ 0.19): every sample has the right mix of grades.

3 Record use the tool and fill in the table

Compare the methods (μ = 7.32 hours)

Use 200 or more samples per row. Values vary a little; the key shows long-run values.

Method	n	Center of $\bar{x}$	SD of $\bar{x}$	Biased?
Simple random sample	16	≈ 7.32	≈ 0.19	no
Stratified by grade	16	≈ 7.32	≈ 0.145	no
Cluster (2 homerooms)	16	≈ 7.32	≈ 0.39	no
Convenience	16	6.66	0	yes (low)
Simple random sample	64	≈ 7.32	≈ 0.08	no

4 Explain what did you discover?

1. A method is biased when its estimates are centered away from (not on) the true value.
2. Larger samples reduce variability but not bias.
3. Stratifying by a variable related to the response reduces variability.
4. Cluster sampling works best when each cluster looks like a small version of the population.

5. Use the lab's results to explain the difference between bias and variability.
   Bias is about where the estimates center: convenience samples center far below μ. Variability is about how spread out they are: stratified samples spread less than SRSs, and single-grade homeroom clusters spread more.

5 Apply a new situation, no tool

1. A school's four grades have very different sleep habits. To estimate average sleep with 80 students, which is better: an SRS of 80 or 20 random students from each grade? Why?
   Stratified (20 per grade): grade is related to sleep, so stratifying reduces variability; both are unbiased.
2. Identify the method: choose 5 of the school's 40 homerooms at random and survey everyone in them.
   Cluster sampling.
3. A news website poll gets 12,000 responses. Is the result trustworthy? Explain.
   No: voluntary response bias. People with strong opinions choose to answer, and a large sample does not fix bias.

Teacher notes

Intended discovery: Bias is where a method's estimates center; variability is how much they spread. Random methods are unbiased, larger n reduces variability but not bias, stratifying on a related variable reduces variability, and clusters that resemble strata increase it.

Likely misconceptions:

- A random sample is guaranteed to be representative.
- A huge sample is accurate regardless of how it was chosen.
- Confusing stratified and cluster sampling.

Answers that can vary: Explore 1 values and all simulated centers and SDs (the key gives long-run values from 100,000 simulated samples).